

Streaming flow — SRT / NDI

Primary path (SRT)

```
VMix PCs (both nodes) --SRT--> Restreamer (mothership) --SRT (fan-out)--> BirdDog Play x12 (one per theatre)
```

SRT is unicast-by-IP, so this path has no discovery dependency and should route cleanly over the tailnet.

Backup path (NDI)

```
BirdDog Central (mothership) --NDI--> BirdDog Play x12 (one per theatre)
```

How Central gets the program — NDI redistribution (settled). Central works as an NDI *redistributor*: the source VMix node PC's native NDI output registers with the NDI Discovery Server, Central receives it over that node's own uplink (~130 Mbps of the node's ~170 Mbps ceiling, a **fallback-only** load — pausing that node's VMix record ingest for the duration keeps it under the ceiling, the same lever the theatre side already uses), and Central re-sends it to the theatre PLAYS. Two researched facts pin this shape:

- **BirdDog Central is NDI-only on the input side** — its own user guide lists NDI sources, an NDI file-player, and an NDI re-transmitter; there is no SRT/RTMP ingest anywhere in it (Central 2.0 User Guide). So the feed into Central must already be NDI — it can't terminate the SRT program.
- **Restreamer can't be the NDI sender instead** — upstream FFmpeg removed NDI support in 2019 over a NewTek GPL violation and never reinstated it; datarhei's own FFmpeg build contains no NDI library and the maintainers have declined to ship it for licensing reasons (datarhei/restreamer discussion #391).

Remaining verification items (not design questions) are tracked as `docs/open-questions.md` #16. BirdDog Central was deliberately retained for this sender role when its fleet-management duties moved to Flock — see `docs/birddog-play-rationale.md`.

Bandwidth note: full NDI (not HX) at 1080p50 is ~125 Mbps per stream, roughly constant regardless of content — 12-22x SRT's figures, and without SRT's "static content is cheap" discount. Fine for a single theatre falling back — *provided that theatre's ATEM ISO ingest is paused for the duration*, since NDI plus the ingest together exceed the theatre's own A-1300 WireGuard ceiling — and a real capacity question if this path ever has to carry all 12 theatres at once (e.g. a Restreamer failure). See `docs/bandwidth-analysis.md` for the full impact at per-router, per-VLAN, and mothership-NIC level.

Caveat, and the fix: NDI relies on mDNS multicast for discovery, which does not cross a Tailscale-routed boundary — Tailscale is point-to-point WireGuard and doesn't forward multicast/mDNS (still an open feature

request, not something MagicDNS papers over). Plain local mDNS discovery will not find BirdDog Central across theatres on its own.

Plan: run an NDI Discovery Server on the mothership. NDI 5+ added this specifically for WAN/VPN scenarios — it's unicast TCP (default port **5959**), so it works fine across the tailnet with no multicast involved:

- Run the discovery server as a small service on the mothership, alongside BirdDog Central.
- Point every NDI participant at that server's IP: the VMix node PCs (senders of the program feed Central redistributes), BirdDog Central (receiver of that feed *and* sender toward the theatres), and all 12 BirdDog Play units (receivers). On Windows/macOS this is the NDI Tools Access Manager "Advanced" tab; on Linux it's `~/ndi/ndi-config.v1.json`.
- Once a sender has a discovery server configured, it stops using mDNS entirely — it's visible only to finders pointed at the same server.
- Tailscale MagicDNS can supply a stable hostname for the discovery server's address instead of hardcoding its Tailscale IP, if the receiving device's config field accepts a hostname — a convenience, not a substitute for the discovery server itself.

Confirmed on BirdDog Play: BirdUI (the PLAY's web admin — browse to its IP, default password `birddog`) has a Network panel with NDI-specific settings including a Discovery Server toggle. Switch it ON, enter a comma-delimited list of Discovery Server IP address(es), click APPLY. BirdDog's own docs describe this as the mechanism for locating NDI sources "on different subnets" — exactly this cross-theatre case — and it supports multiple servers for failover. Sources: BirdDog PLAY User Guide, BirdUI User Guide. Change the default admin password before the event — BirdUI grants full device config access.

See `diagrams/streaming-flow.svg` — primary path solid, backup path dashed. For why BirdDog Play is the theatre-side receiver at all (vs. a laptop

- VLC), see `docs/birddog-play-rationale.md`.